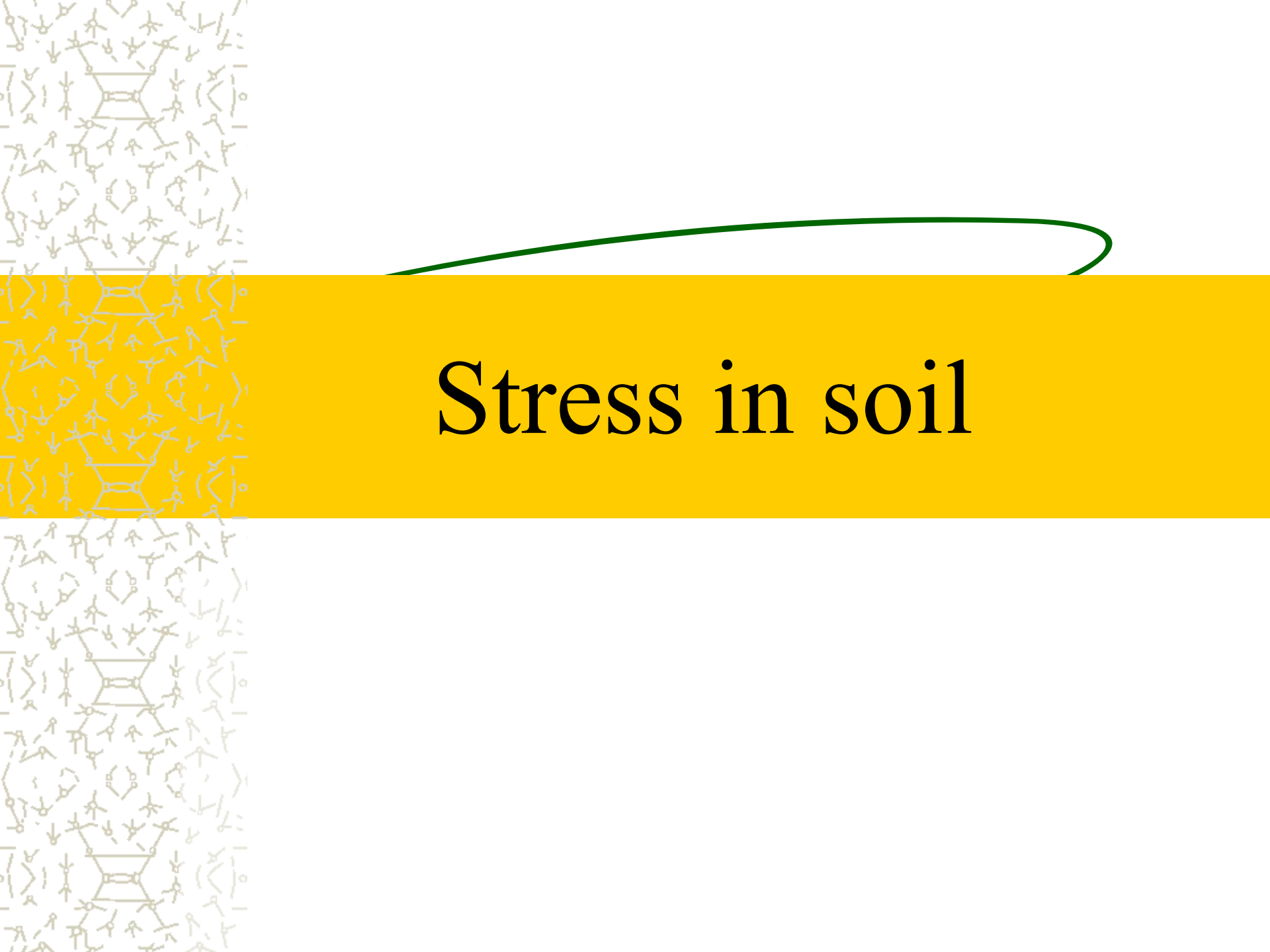
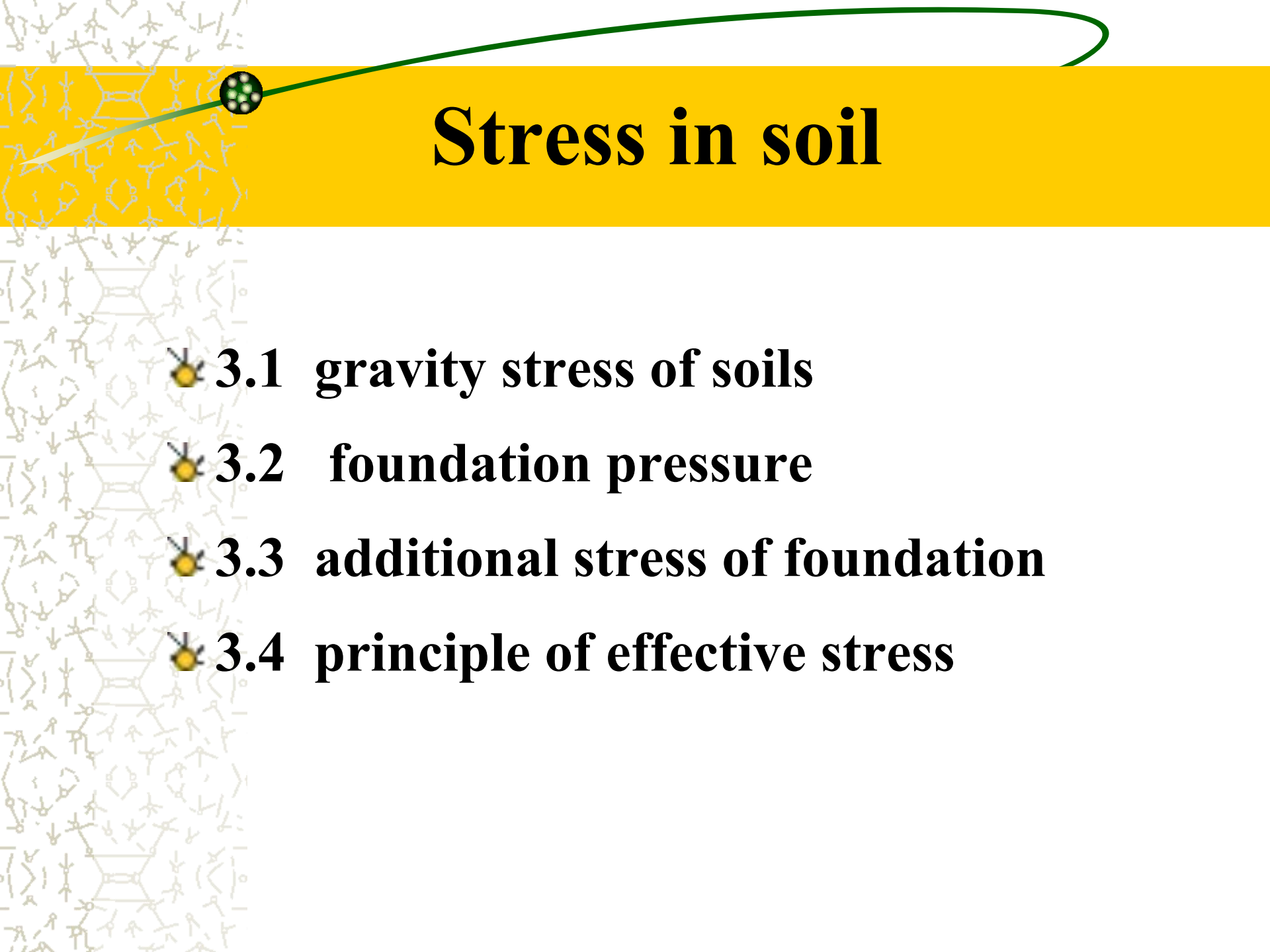


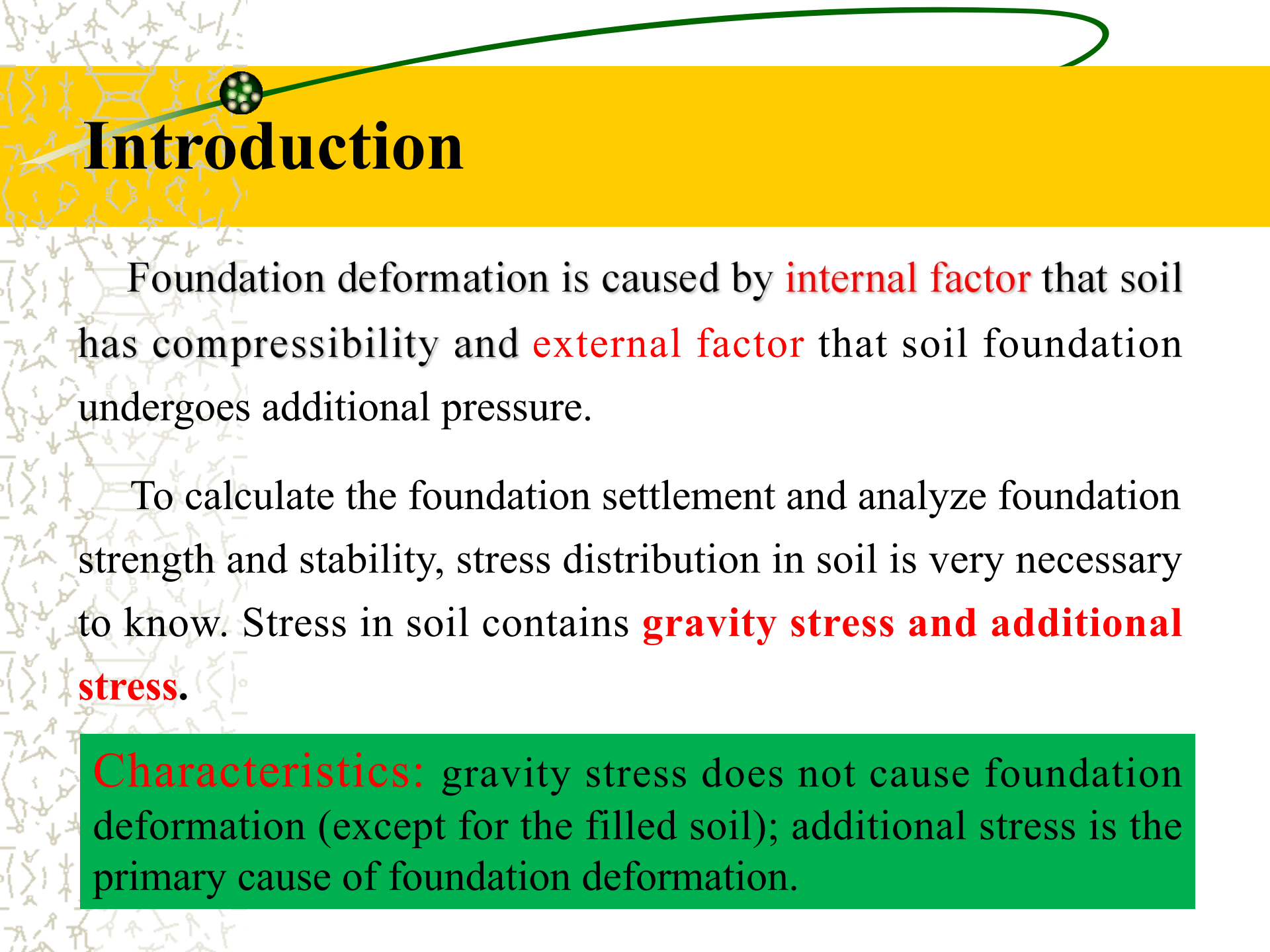


Stress in soil



Stress in soil

- **3.1 gravity stress of soils**
- **3.2 foundation pressure**
- **3.3 additional stress of foundation**
- **3.4 principle of effective stress**



Introduction

Foundation deformation is caused by **internal factor** that soil has compressibility and **external factor** that soil foundation undergoes additional pressure.

To calculate the foundation settlement and analyze foundation strength and stability, stress distribution in soil is very necessary to know. Stress in soil contains **gravity stress and additional stress**.

Characteristics: gravity stress does not cause foundation deformation (except for the filled soil); additional stress is the primary cause of foundation deformation.

Basic assumption

- ✦ Foundation soil is isotropy, homogenous, linear elasticity.
- ✦ Foundation soil is limitless in depth and horizontal direction

Foundation: homogeneous isotropic linearly deformed
semi-space body

To solve elastic half-space problems
with the **theory of elasticity**

A decorative header featuring a solid yellow horizontal bar. A green line starts from the left edge of the bar, curves upwards and to the right, then loops back to the left, ending at a small black circle with a green dot pattern. The left side of the slide has a vertical strip with a repeating pattern of stylized, interconnected geometric shapes in grey and white.

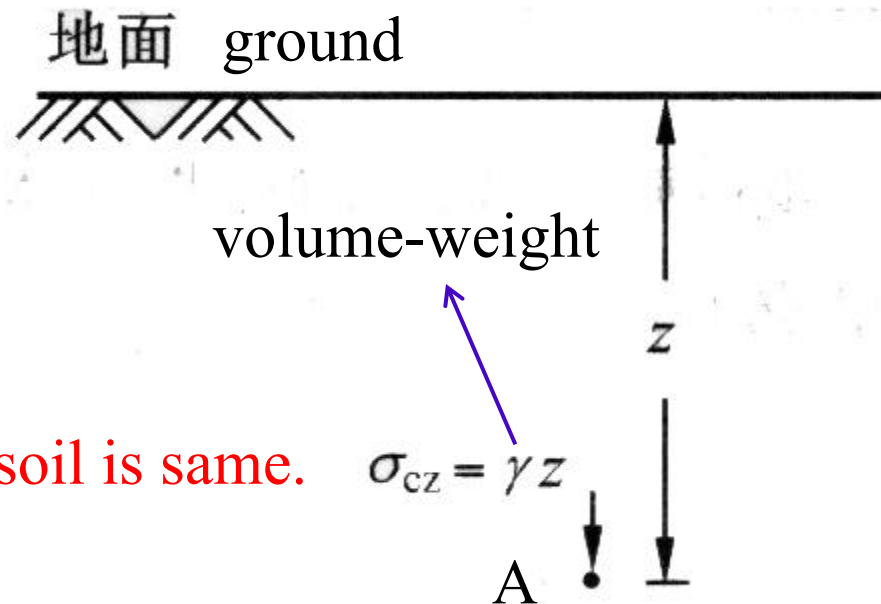
3.1 Gravity stress of soil

Gravity stress of foundation soil

Note: only **intergranular stress** can make the soil particles compacted each other, and causes soil deformation. Moreover, intergranular stress is a key factor to affect soil strength, so it is also called **effective stress**.

Thus, gravity stress of soil is defined as a stress caused by self effective gravity of soil. Vertical and horizontal gravity stresses are refer to **effective gravity stress**. For convenience, σ_{CZ} is usually used to show gravity stress, and σ_C is effective stress.

3.1.1 Gravity stress of homogeneous soil in vertical direction

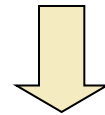


The density of soil is same.

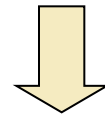
Gravity stress of soil can be calculated by the effective weight of soil per unit area of a point.

3.1.2 Gravity stress of homogeneous soil in horizontal direction

$$\varepsilon_x = \frac{\sigma_{cx}}{E} - \frac{\nu}{E} (\sigma_{cz} + \sigma_{cy}) = 0$$



$$\sigma_{cx} = \sigma_{cy} = \frac{\nu}{1-\nu} \sigma_{cz}$$



$$\sigma_{ch} = \sigma_{cx} = \sigma_{cy} = K_0 \sigma_{cz}$$

3.1.2 Gravity stress of homogeneous soil in horizontal direction

✨ K_0 ——coefficient of lateral earth pressure, it is the ratio of horizontal pressure to vertical stress in uniaxial deformation direction.

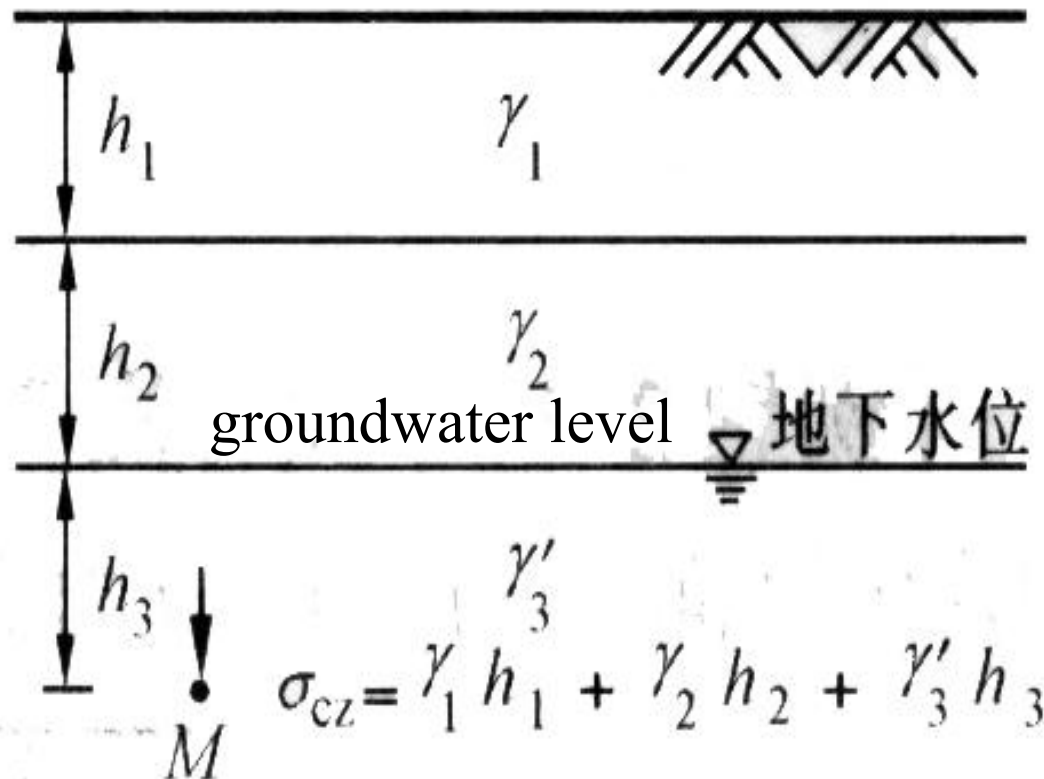
✨ The value can be determined by test, and it is relate to the stress history of soils, soil type and unit weight of soil, et al.

Typical value of K_0

Soil type	K_0
loose sand	0.4-0.45
dense sand	0.45-0.5
compacted fill	0.8-1.5
normally consolidated clay	0.5-0.6
over consolidated soil	1.0-4.0

3.1.3 Gravity stress of layered soils

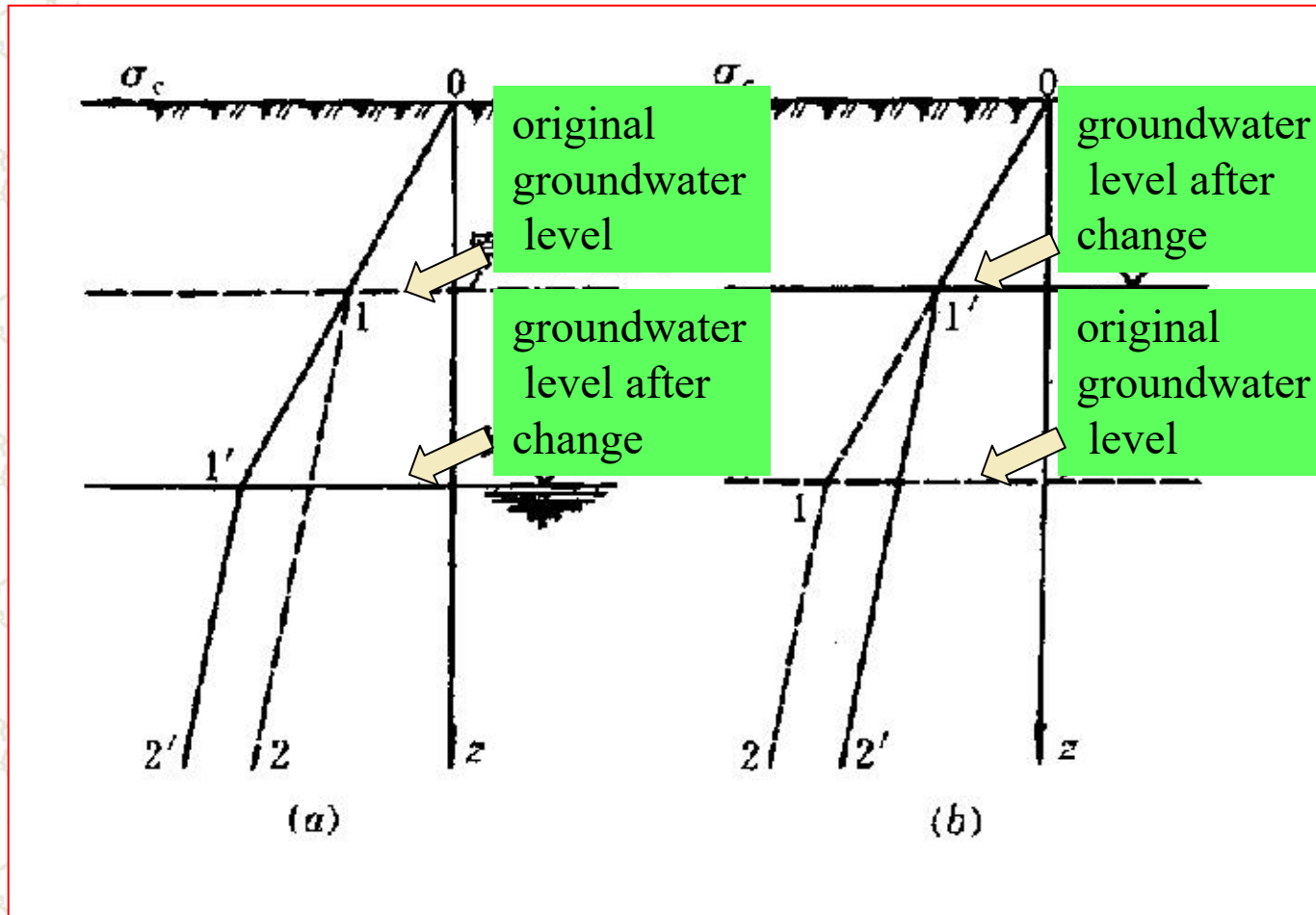
Each layered soil with different density



3.1.3 Gravity stress of layered soils

$$\sigma_{cz} = \sigma_c = \sum_{i=1}^n \gamma_i h_i$$

3.1.4 effect of groundwater level rise and fall





Law of distribution curve of gravity stress

- ✿ Distribution curve of gravity stress of soil is a broken line, inflection point is at soil boundary or groundwater level
- ✿ Gravity stress of soil at same layer changes in a linear fashion.
- ✿ Gravity stress of soil increases as depth increases.

【example 3-1】 calculate gravity stress and draw distribution of gravity stress

bar graph depth layer thickness calculation of gravity stress

soil
type

silt

粉

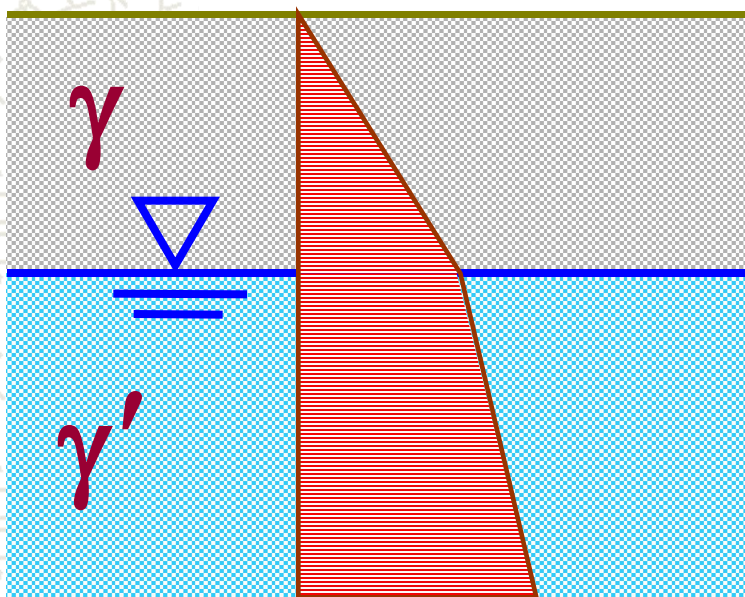
土

silt

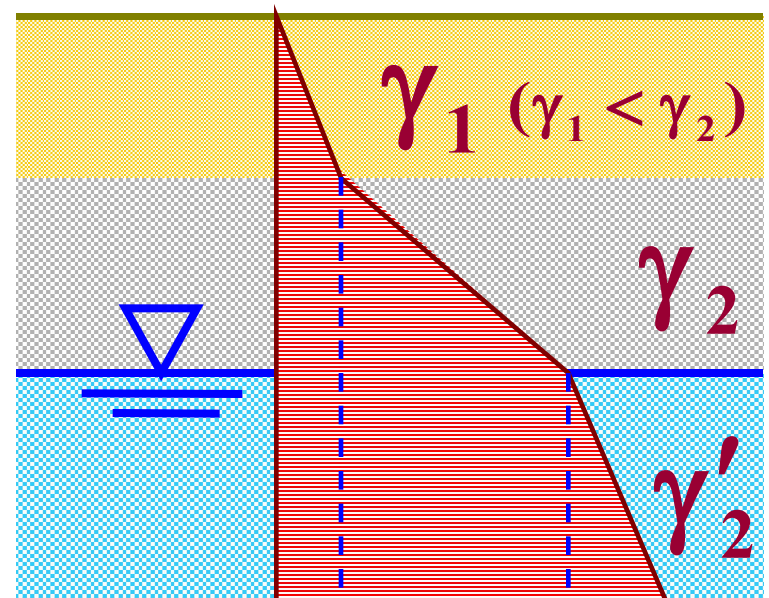
clay

粉质粘土

土层	土的有效重度的计算	柱状图	深度 z (m)	分层厚度 h_i (m)	重度 γ_i (kN/m ³)	竖向自重应力计算 σ_c (kPa)	竖向自重应力分布图 distribution diagram
粉土	$\begin{cases} \gamma = 18.0 \text{ kN/m}^3 \\ d_s = 2.70 \\ w = 35\% \end{cases}$ $\gamma' = \frac{(d_s - 1)\gamma_w}{1 + e} \text{ 地下水位}$ $= \frac{(d_s - 1)\gamma}{d_s(1 + w)}$ $= \frac{(2.70 - 1) \times 18.0}{2.70 \times (1 + 0.35)}$ $= 8.4 \text{ kN/m}^3$		2.5			$18 \times 2.5 = 45$	
			3.6	3.6	18	$18 \times 3.6 = 65$	
			5.0			$65 + 8.4(5 - 3.6) = 77$	
			6.0	2.4	8.4	$65 + 8.4(6 - 3.6) = 85$	
粉质粘土	$\begin{aligned} \gamma &= 18.9 \text{ kN/m}^3 \\ d_s &= 2.72 \\ w &= 34.3\% \end{aligned}$ $\gamma' = \frac{(2.72 - 1) \times 18.9}{2.72 \times (1 + 0.343)}$ $= 8.9 \text{ kN/m}^3$		9.0		8.9	$85 + 8.9(9 - 6) = 112$	112kPa



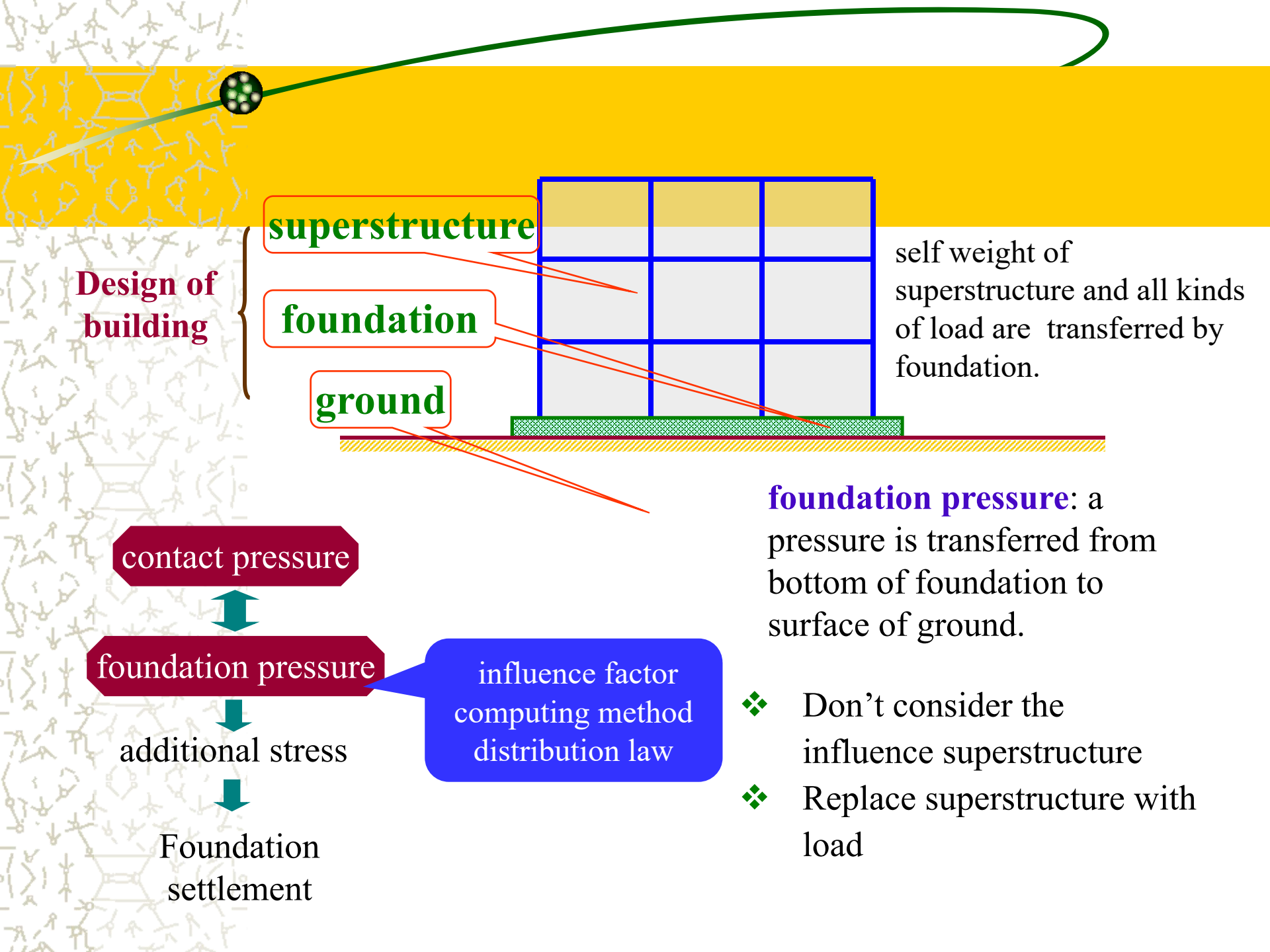
homogeneous foundation



layered foundation



3.2 foundation pressure and additional stress of foundation

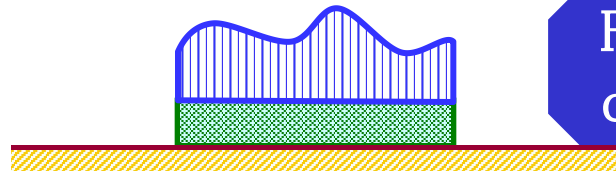


1. Influence factors

**Foundation
stress**

magnitude
direction
distribution

loading condition



Foundation
condition

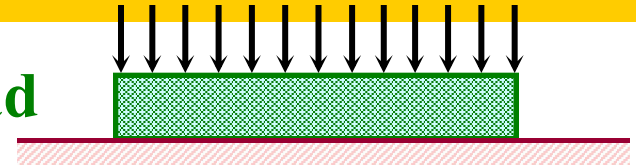
- Rigidity
- shape
- size
- buried depth

ground condition

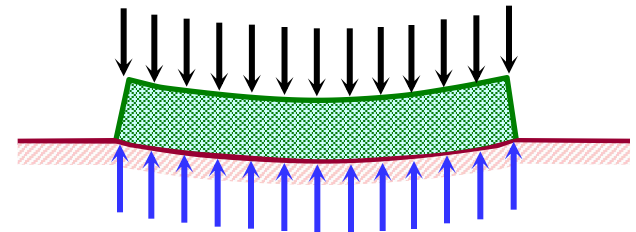
Soil type
density
Soil Layer Structure

2. Distribution of foundation pressure

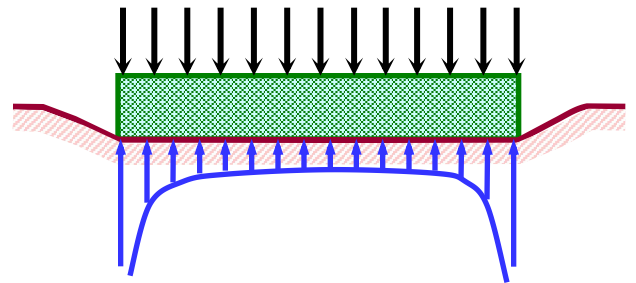
strip foundation, **vertical distributed load**



elastic foundation, *flexible foundation*



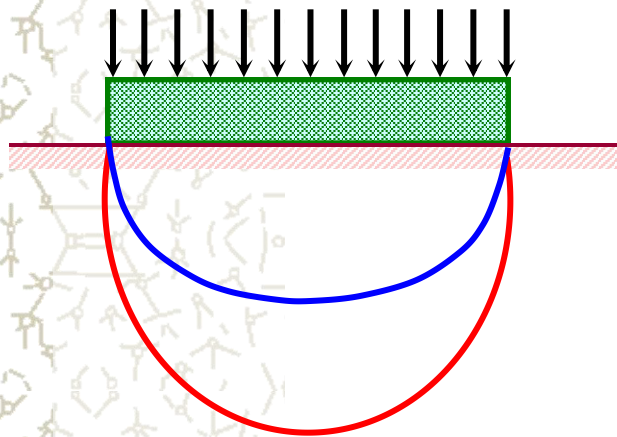
elastic foundation ,
absolutely rigid foundation



contact pressure distribution

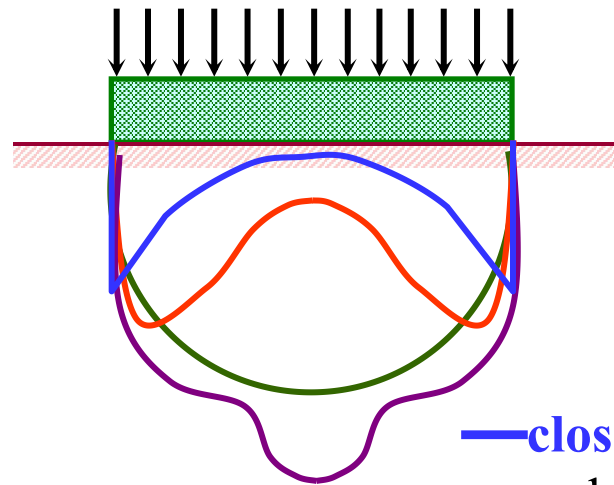
Distribution of foundation pressure

Elastoplastic foundation, Finite stiffness foundation



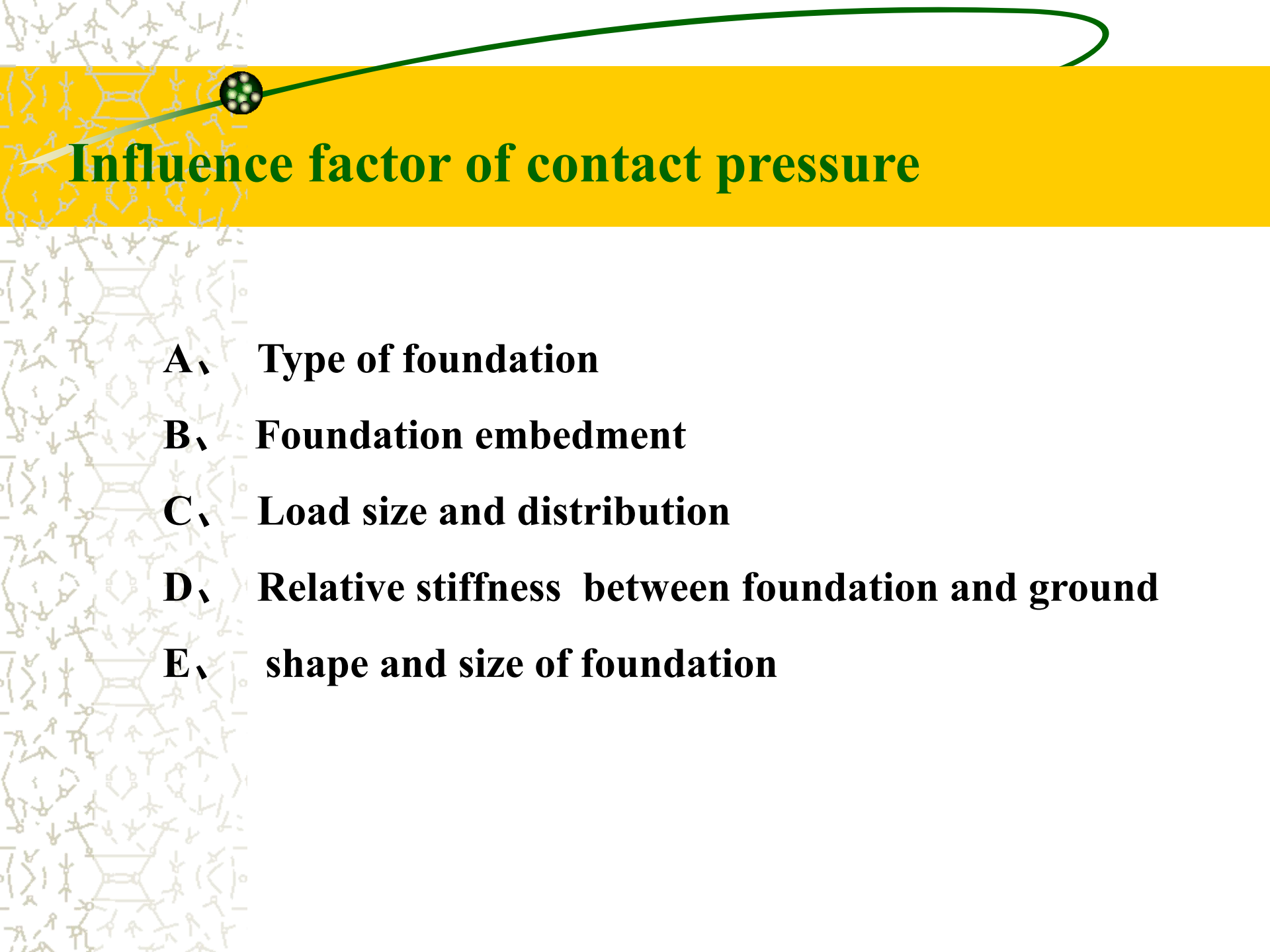
— smaller load
— larger load

Sand ground



— close to elastic solution
— saddle-type
— parabolic-type
— inverted bell shape

Cohesive soil ground



Influence factor of contact pressure

- A、 Type of foundation**
- B、 Foundation embedment**
- C、 Load size and distribution**
- D、 Relative stiffness between foundation and ground**
- E、 shape and size of foundation**

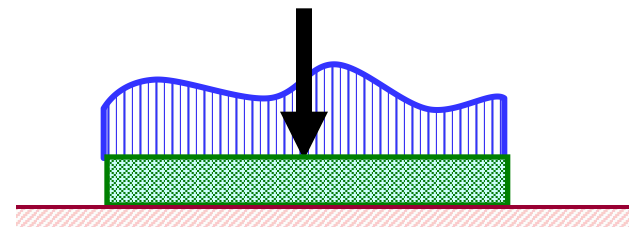
Practical simplified calculation

Distribution
of Contact
pressure is
very
complicated

Based on **Saint-Venant principle**, the influence of the distribution of contact pressure on calculation of foundation stress is **only limited to a certain depth range**. Beyond the range, the distribution of additional stress in ground has little relation with the distribution of contact pressure, additional stress depends on the size, direction and location of load resultant force .

**The size of foundation is
smaller and the load is
not very large.**

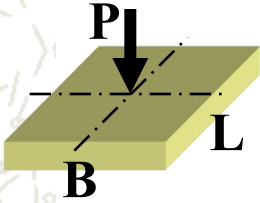
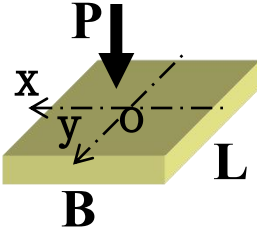
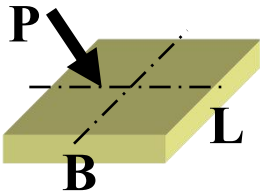
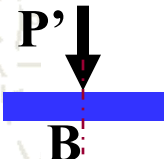
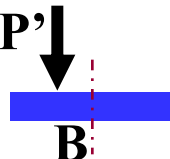
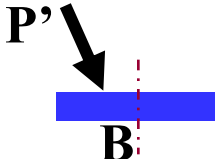
simplified calculating method:
Contact pressure is distributed linearly.



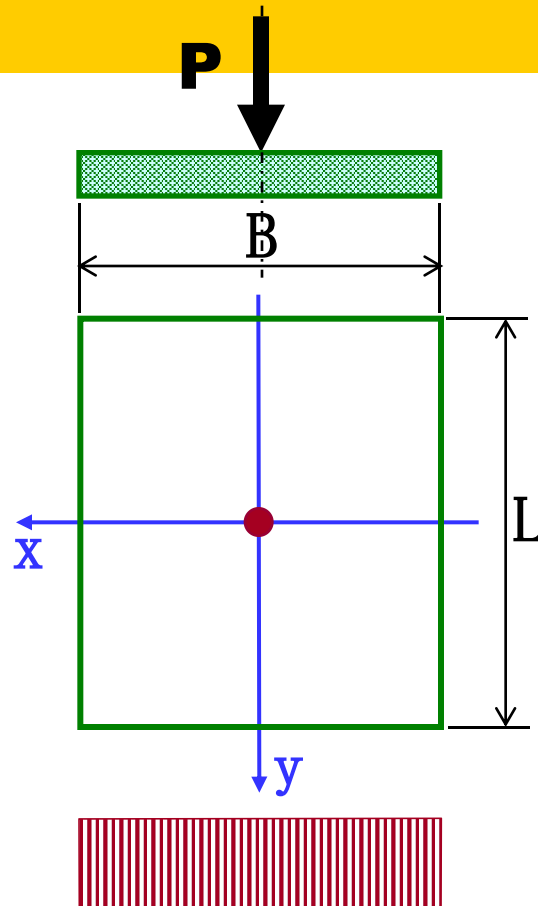
Difference load form

loading condition

Shape of foundation

	vertical center	vertical eccentric	inclined eccentric
rectangle			
strip-type			

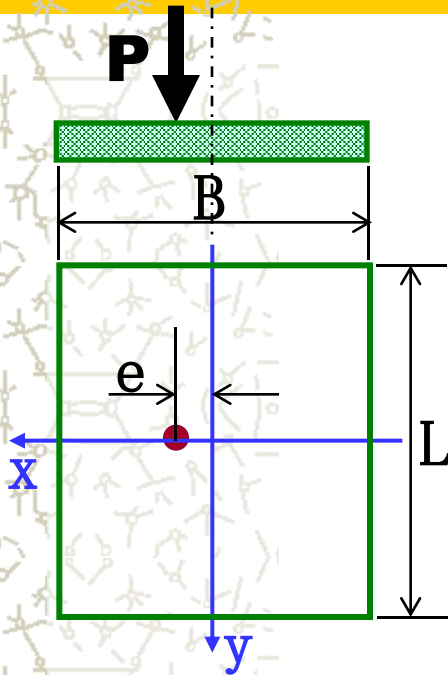
rectangular cross-section centric load



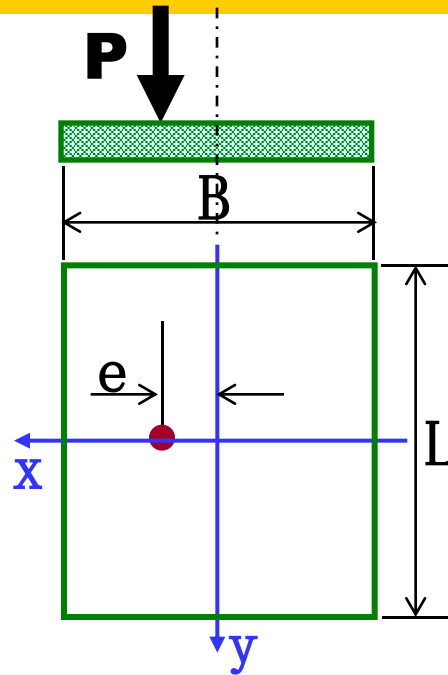
$$p = P/A$$

$$p_{\max/\min} = \frac{P}{A} \pm \frac{M}{W} = \frac{P}{A} \left(1 \pm \frac{6e}{B} \right)$$

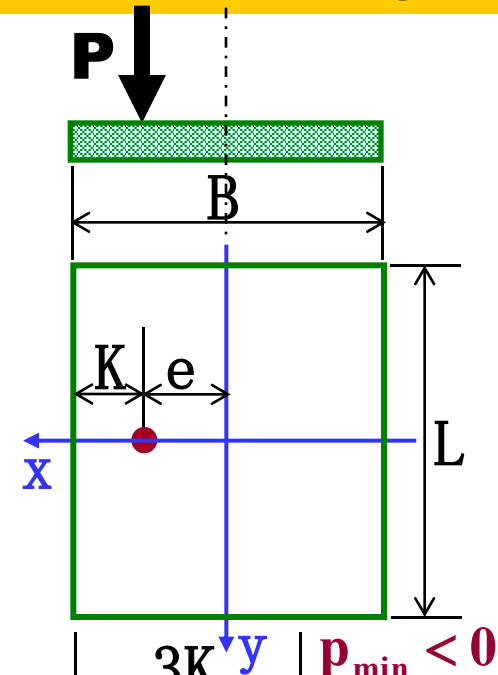
rectangular cross-section eccentric loading



$e < B/6$: trapezoid



$e = B/6$: triangle

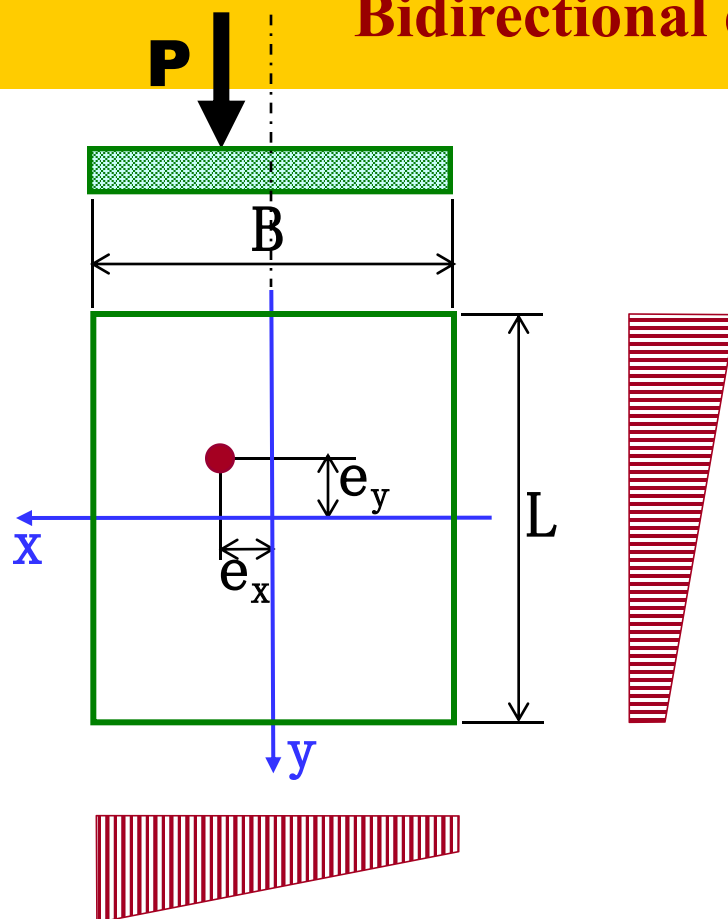


$$p_{\max} = \frac{2P}{3KL} = \frac{2P}{3(B/2 - e)L}$$

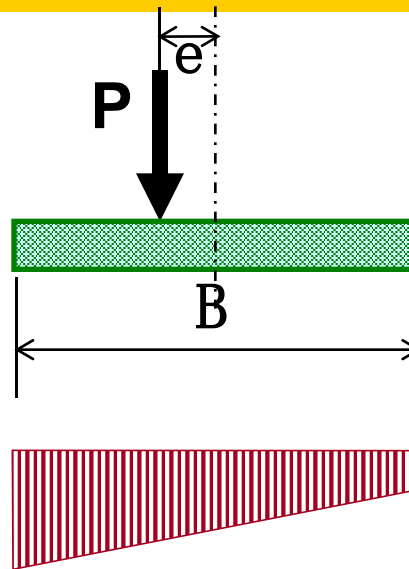
$e > B/6$:

Tensile stress zone appears

Bidirectional eccentric load



Strip foundation vertical eccentric load

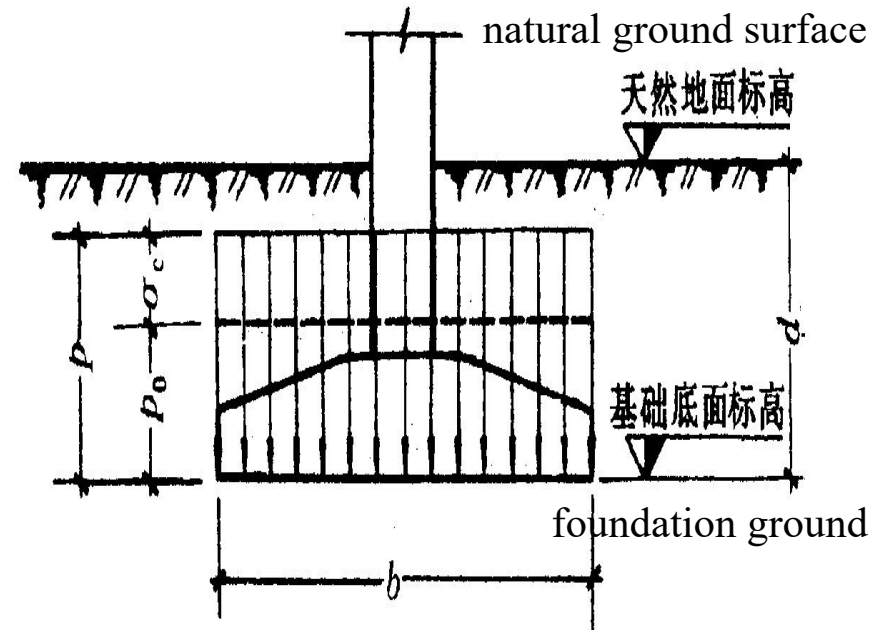


$$p_{\max} = \frac{P'}{B} \left(1 + \frac{6e}{B} \right)$$
$$p_{\min} = \frac{P'}{B} \left(1 - \frac{6e}{B} \right)$$

Additional pressure at foundation bottom—— net pressure

additional pressure at foundation bottom: **a new pressure** on foundation bottom that gravity stress is taken out from contact pressure after construction.

$$p_0 = p - \sigma_c = p - \gamma_0 d$$



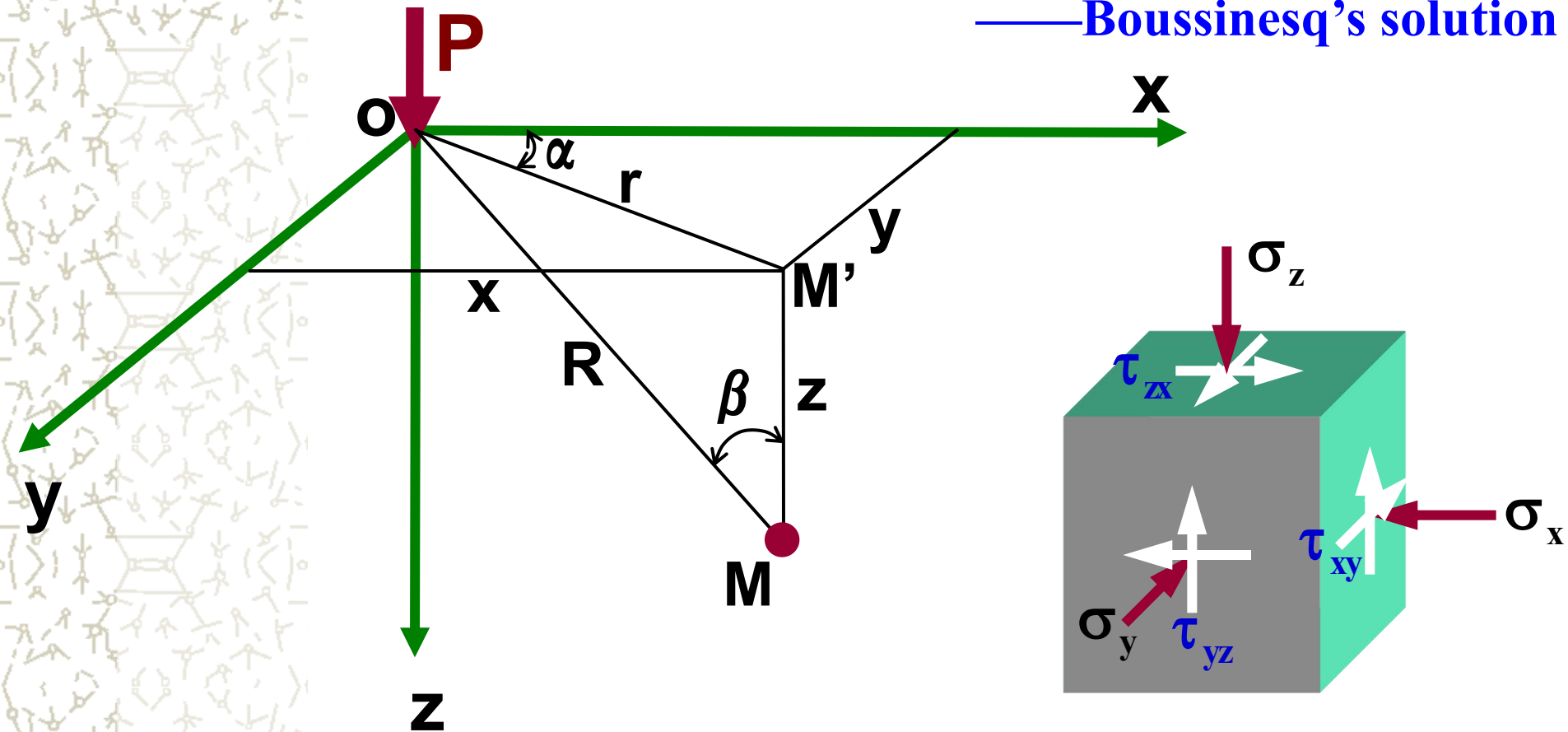
Calculation of additional pressure at
foundation bottom

3.3 Additional foundation pressure

1. **Definition:** a stress in ground is produced by **building load**.
2. **Assumption :** based on **elastic theory**
 - a. ground is taken as semi-space elastomer;
 - b. foundation soil is uniformly continuous; (modulus of deformation E and coefficient of lateral expansion μ **are equal** everywhere)
 - c. foundation soil is isotropic; (E and μ in same point are **equal** in all directions)

Calculation of additional stress under vertically concentrated force action

— Boussinesq's solution



$$R^2 = r^2 + z^2 = x^2 + y^2 + z^2$$

$$r / z = \operatorname{tg} \beta$$

$$\sigma_z = \frac{3P}{2\pi} \frac{z^3}{R^5} \quad \tau_{zy} = \frac{3P}{2\pi} \frac{yz^2}{R^5} \quad \tau_{zx} = \frac{3P}{2\pi} \frac{xz^2}{R^5}$$

$$R^2 = r^2 + z^2 = x^2 + y^2 + z^2$$

$$\sigma_z = \frac{3P}{2\pi} \frac{z^3}{R^5} = \frac{3}{2\pi} \frac{1}{[1 + (r/z)^2]^{5/2}} \frac{P}{z^2}$$

$$\alpha = \frac{3}{2\pi} \frac{1}{[1 + (r/z)^2]^{5/2}}$$

distribution coefficient

Specific derivation
can be found in
elastic mechanics
(Xu zhilong)
pp.220

$$\sigma_z = \alpha \cdot \frac{P}{z^2}$$

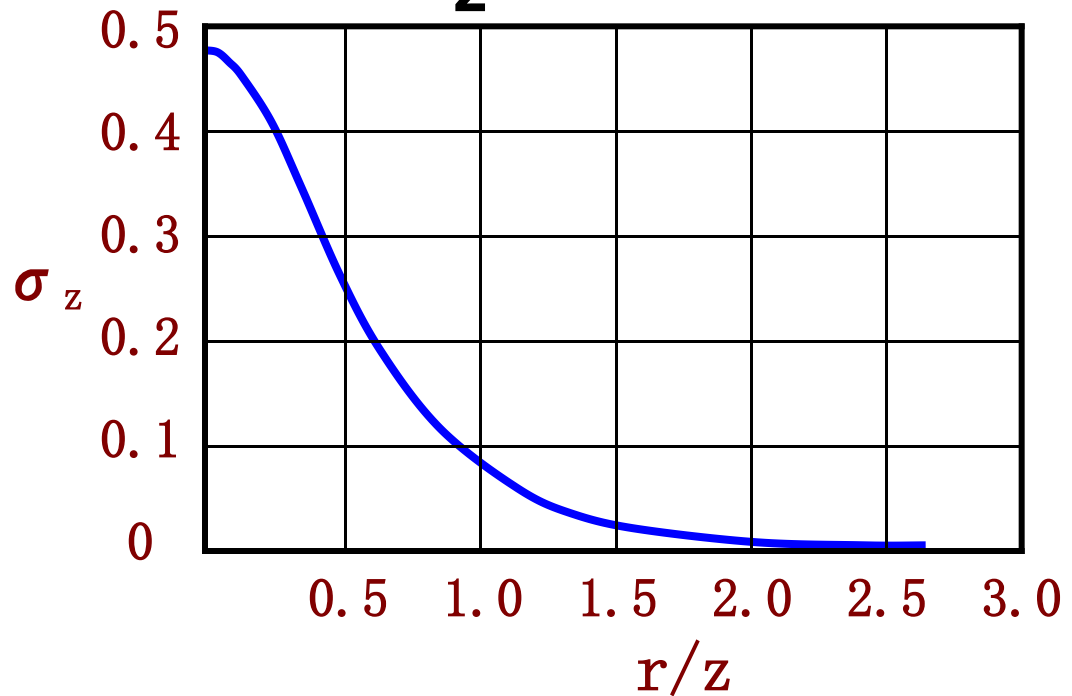
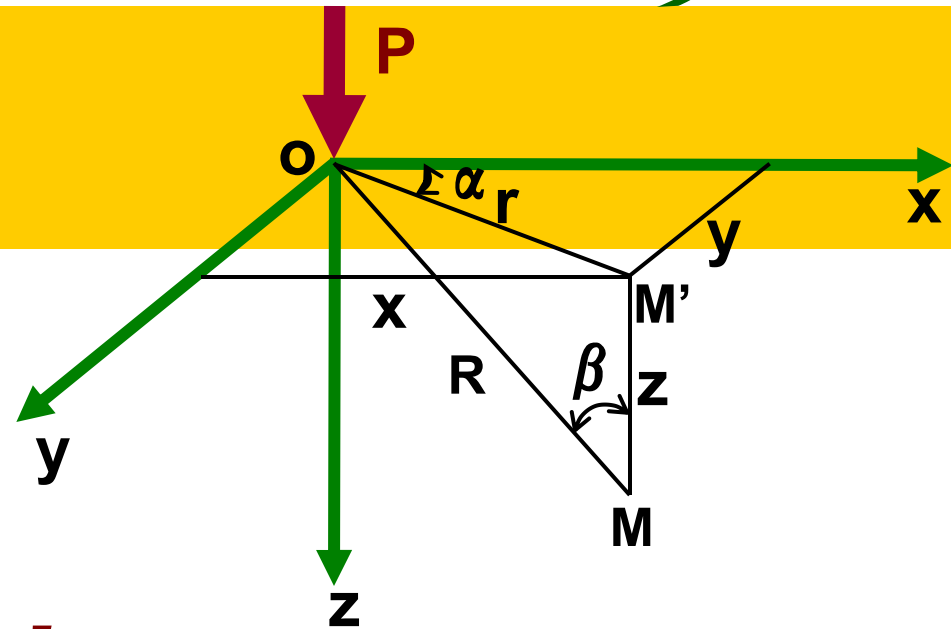
Get from
table 9.1

Characteristic

$$\sigma_z = \alpha \cdot \frac{P}{z^2}$$

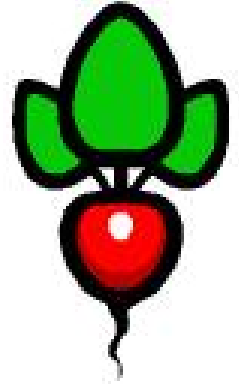
$$\alpha = \frac{3}{2\pi} \frac{1}{[1 + (r/z)^2]^{5/2}}$$

1. The stress is distributed axisymmetrically

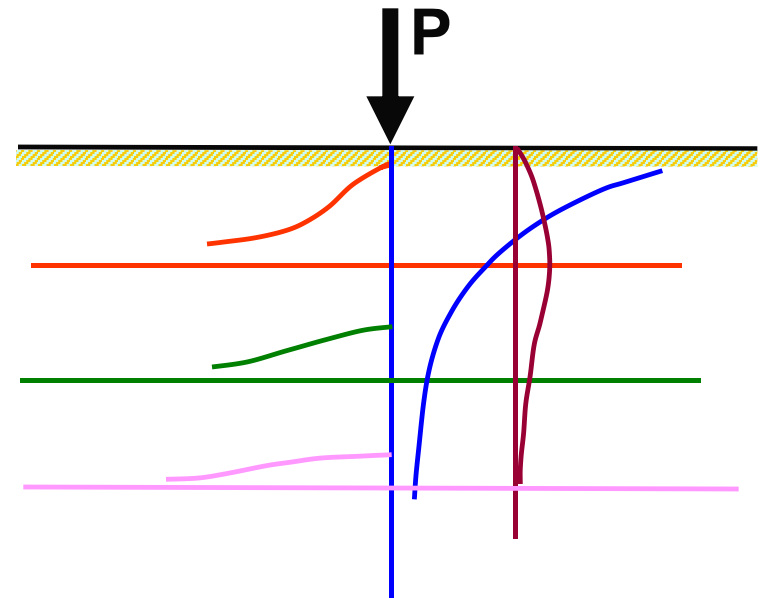
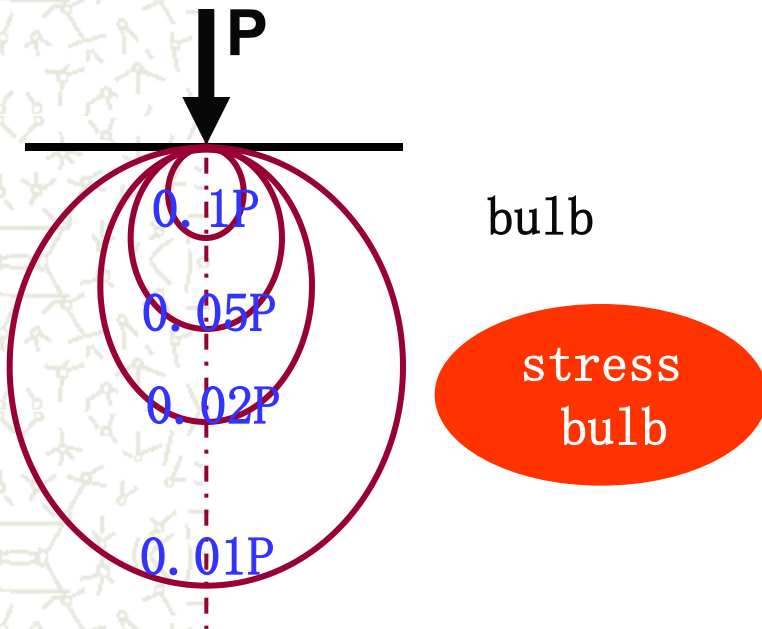


$$\sigma_z = \alpha \cdot \frac{P}{z^2}$$

$$\alpha = \frac{3}{2\pi} \frac{1}{[1 + (r/z)^2]^{5/2}}$$



2. In P direction, $r=0$, $\alpha=3/(2\pi)$, $z=0$, $\sigma_z \rightarrow \infty$, $z \rightarrow \infty$, $\sigma_z=0$
3. At a certain horizontal surface, $z=\text{const}$, $r=0$, α 最大, $r \uparrow$, $\alpha \downarrow$, $\sigma_z \downarrow$
4. at a certain cylindrical surface, $r=\text{const}$, $z=0$, $\sigma_z=0$, $z \uparrow$, σ_z increases first and then decreases
5. σ_z contour line — Stress bubble



Additional stress calculation under uniformly distributed load of rectangular area

$$dP = p dx dy$$

1. additional stress under corner detection

$$d\sigma_z = \frac{3dP}{2\pi} \frac{z^3}{R^5} = \frac{3p}{2\pi} \frac{z^3}{R^5} dx dy$$

$$\sigma_z = \int_0^B \int_0^L d\sigma_z = \sigma_z(p, m, n)$$

$$m = B/z, \quad n = L/z$$

$$\alpha_c = F\left(\frac{B}{z}, \frac{L}{z}\right) = F(m, n)$$

Stress distribution coefficient under a rectangularly loaded area

$$\sigma_z = \alpha_c p$$

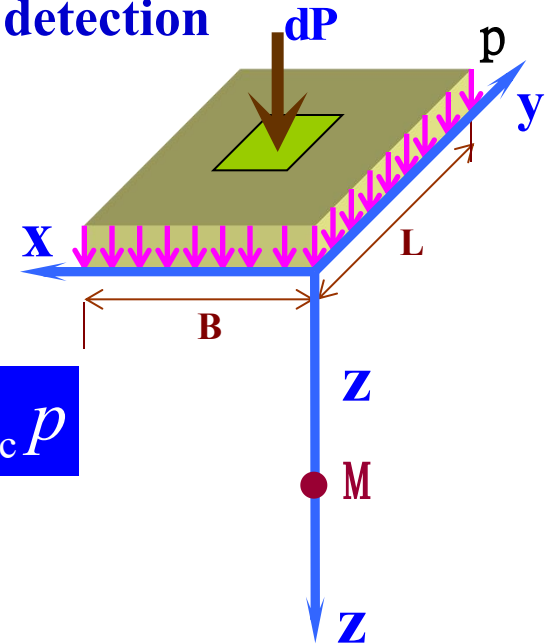


Table 9.31
"Soil Mechanics"

2. vertical additional stress of any point — corner points method

linear relationship
between load and
stress

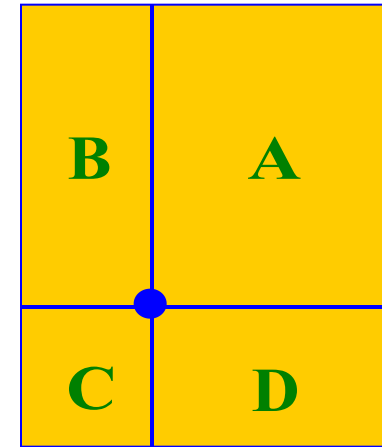


superposition
principle

computational
formula

vertical additional stress of
any point at foundation

corner
points
method



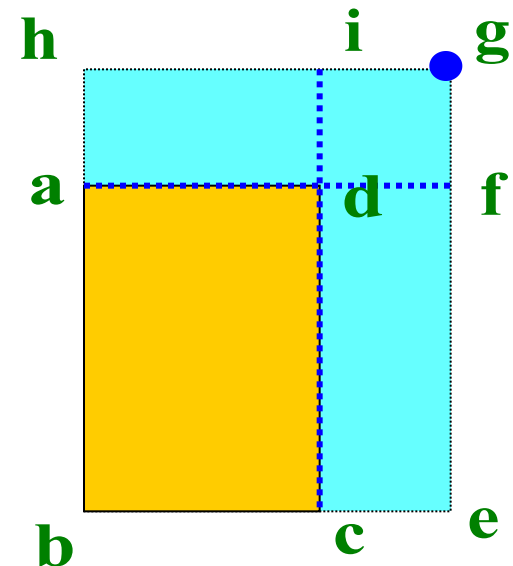
Two situations

a. In rectangular area

$$\sigma_z = (\alpha_c^A + \alpha_c^B + \alpha_c^C + \alpha_c^D) p$$

b. Outside rectangular area

$$\sigma_z = (\alpha_c^{begh} - \alpha_c^{afgh} - \alpha_c^{cegi} + \alpha_c^{dfgi}) p$$



Additional stress calculation under triangular distribution load of rectangular area

$$\sigma_z = \int_0^B \int_0^L d\sigma_z = \sigma_z(p_0, m, n)$$

$$\sigma_z = \alpha_t p_0$$

$$\alpha_t = F(B, L, z) = F\left(\frac{B}{z}, \frac{L}{z}\right) = F(m, n)$$

Stress distribution coefficient

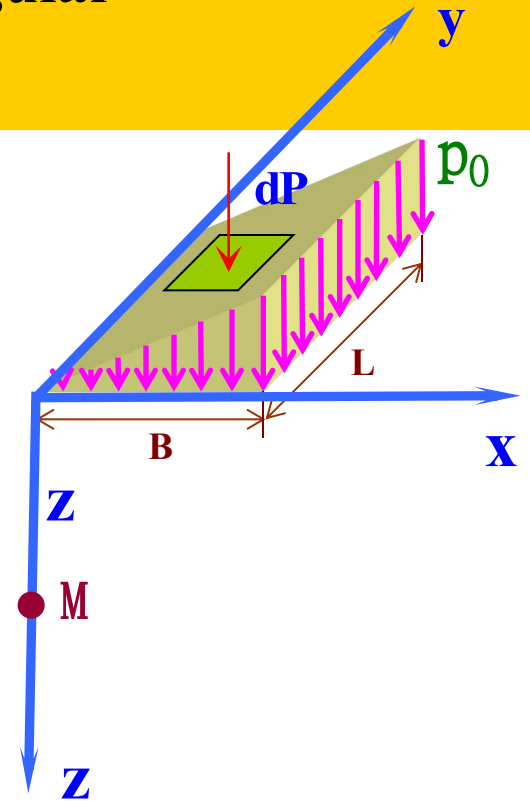


Table 3.6
“soil mechanics and
foundation engineering”
Zhao, 2014

Vertical stress at any point below a uniformly load circular area

$$\sigma_z = \alpha_0 p$$

$$\alpha_0 = F(r / z)$$

r — the radius of the circular area

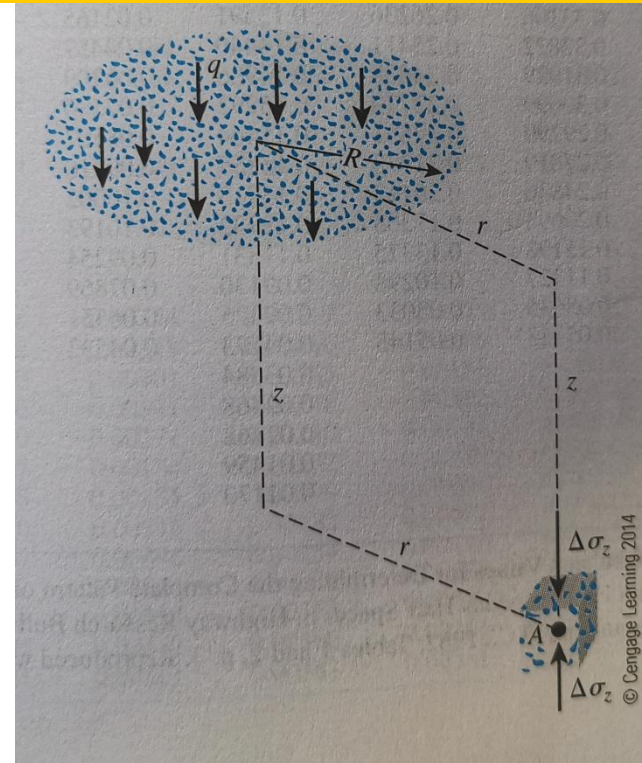


Table 3.6
“soil mechanics and foundation
engineering” Zhao, 2014

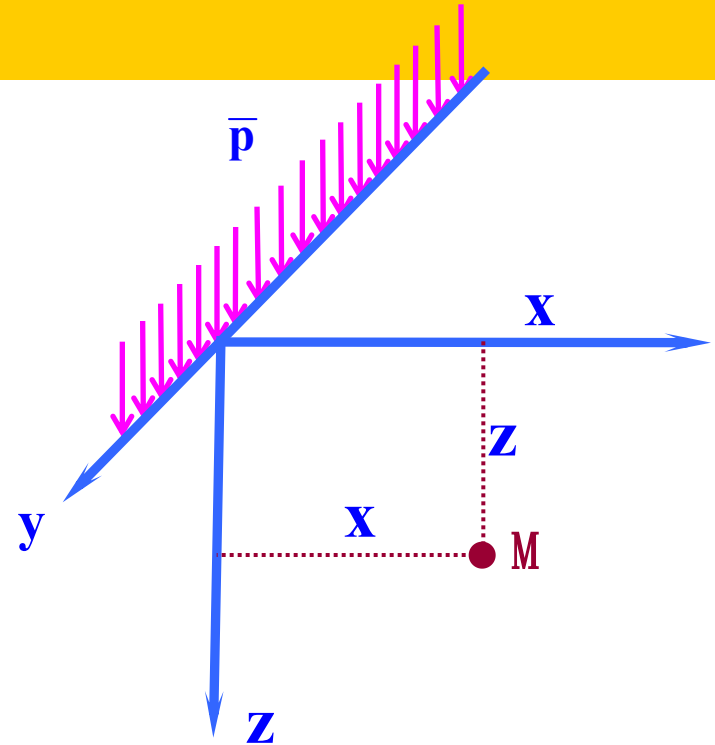
Vertical stress caused by a vertical line load

$$\sigma_z = \frac{2\bar{p}z^3}{\pi(x^2 + z^2)^2}$$

$$\sigma_x = \frac{2\bar{p}x^2z}{\pi(x^2 + z^2)^2}$$

$$\tau_{zx} = \frac{2\bar{p}xz^2}{\pi(x^2 + z^2)^2}$$

$$\sigma_y = \mu(\sigma_x + \sigma_z)$$



Vertical stress caused by a vertical strip load

$$\sigma_z = \alpha_z^s p$$

$$\sigma_x = \alpha_x^s p$$

$$\tau_{xz} = \alpha_{xz}^s p$$

$$\alpha_z^s, \alpha_x^s, \alpha_{xz}^s = F(B, x, z) = F\left(\frac{x}{B}, \frac{z}{B}\right) = F(m, n)$$

Stress distribution coefficient

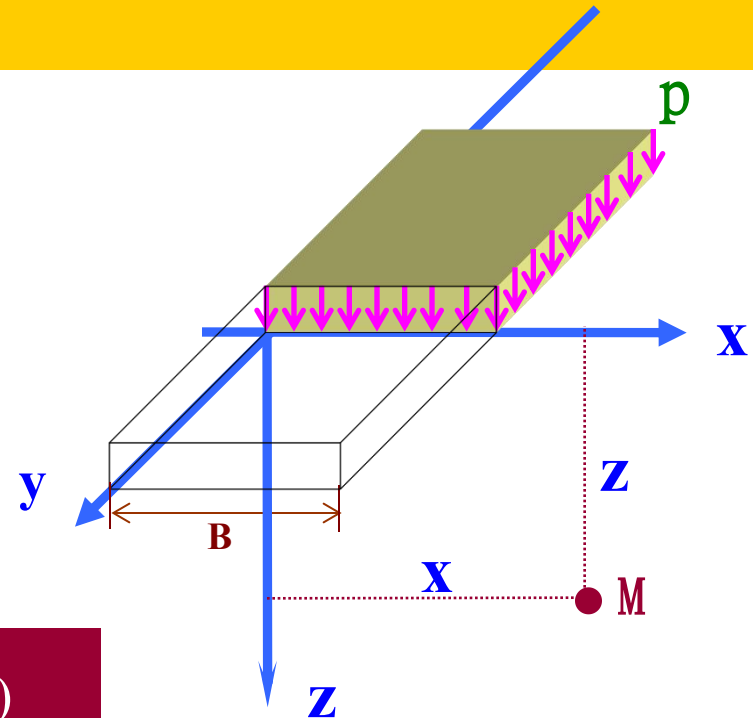
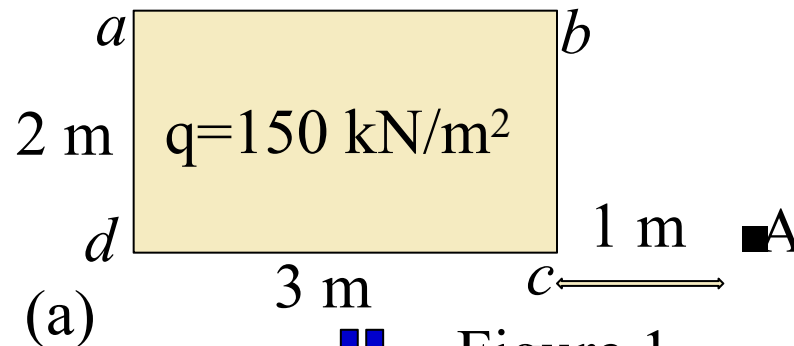


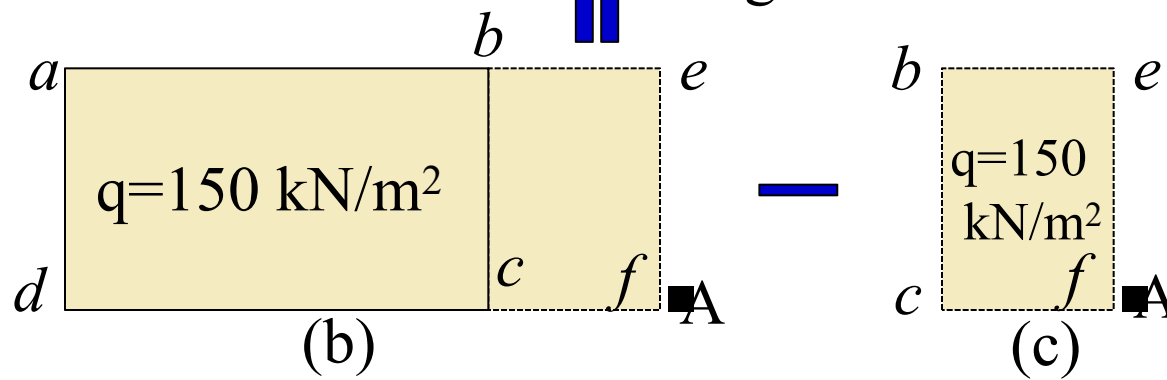
Table 9.4

Example

✱ The plan of a uniformly loaded rectangular area is shown in figure 1. Determine the **vertical stress increases** below point **A** at a depth $z=4$ m.



reminder



Calculation

Solution:

$$\sigma_z = (\alpha_c^{aefd} - \alpha_c^{befc})q$$

For the load area shown in figure 1b:

$$m = \frac{B}{z} = \frac{2}{4} = 0.5, n = \frac{L}{z} = \frac{4}{4} = 1$$

From table 9.31 for $m=0.5$ and $n=1$, the value of $\alpha_c^{aefd}=0.1202$.

Similarly, for the load area shown in figure 1c:

$$m = \frac{B}{z} = \frac{1}{4} = 0.25, n = \frac{L}{z} = \frac{2}{4} = 0.5$$

Thus, $\alpha_c^{befc}=0.0478$.

$$\sigma_z = (\alpha_c^{aefd} - \alpha_c^{befc})q = (0.1202 - 0.0478) \times 150 = 10.86 \text{ kN} / \text{m}^2$$

Table 9.9 Variation of I_3 with m and n [Eq. (9.31)]

n	m									
	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
0.1	0.0047	0.0092	0.0132	0.0168	0.0198	0.0222	0.0242	0.0258	0.0270	0.0279
0.2	0.0092	0.0179	0.0259	0.0328	0.0387	0.0435	0.0474	0.0504	0.0528	0.0547
0.3	0.0132	0.0259	0.0374	0.0474	0.0559	0.0629	0.0686	0.0731	0.0766	0.0794
0.4	0.0168	0.0328	0.0474	0.0602	0.0711	0.0801	0.0873	0.0931	0.0977	0.1013
0.5	0.0198	0.0387	0.0559	0.0711	0.0840	0.0947	0.1034	0.1104	0.1158	0.1202
0.6	0.0222	0.0435	0.0629	0.0801	0.0947	0.1069	0.1168	0.1247	0.1311	0.1361
0.7	0.0242	0.0474	0.0686	0.0873	0.1034	0.1169	0.1277	0.1365	0.1436	0.1491
0.8	0.0258	0.0504	0.0731	0.0931	0.1104	0.1247	0.1365	0.1461	0.1537	0.1598
0.9	0.0270	0.0528	0.0766	0.0977	0.1158	0.1311	0.1436	0.1537	0.1619	0.1684
1.0	0.0279	0.0547	0.0794	0.1013	0.1202	0.1361	0.1491	0.1598	0.1684	0.1752
1.2	0.0293	0.0573	0.0832	0.1063	0.1263	0.1431	0.1570	0.1684	0.1777	0.1851
1.4	0.0301	0.0589	0.0856	0.1094	0.1300	0.1475	0.1620	0.1739	0.1836	0.1914
1.6	0.0306	0.0599	0.0871	0.1114	0.1324	0.1503	0.1652	0.1774	0.1874	0.1955
1.8	0.0309	0.0606	0.0880	0.1126	0.1340	0.1521	0.1672	0.1797	0.1899	0.1981
2.0	0.0311	0.0610	0.0887	0.1134	0.1350	0.1533	0.1686	0.1812	0.1915	0.1999
2.5	0.0314	0.0616	0.0895	0.1145	0.1363	0.1548	0.1704	0.1832	0.1938	0.2024
3.0	0.0315	0.0618	0.0898	0.1150	0.1368	0.1555	0.1711	0.1841	0.1947	0.2034
4.0	0.0316	0.0619	0.0901	0.1153	0.1372	0.1560	0.1717	0.1847	0.1954	0.2042
5.0	0.0316	0.0620	0.0901	0.1154	0.1374	0.1561	0.1719	0.1849	0.1956	0.2044
6.0	0.0316	0.0620	0.0902	0.1154	0.1374	0.1562	0.1719	0.1850	0.1957	0.2045

Factors affecting stress distribution in soil

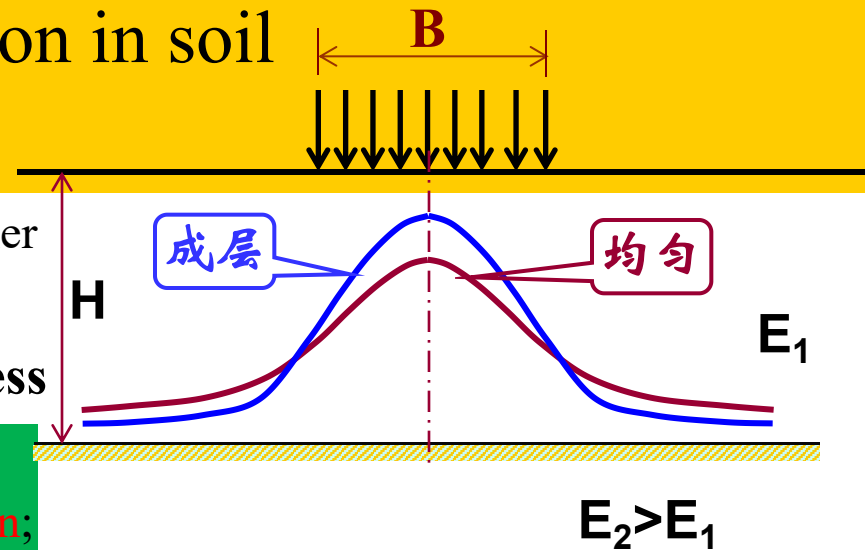
1. Nonlinearity and elastoplasticity

The influence is greater when the stress level is higher

2. Heterogeneity—layered soil

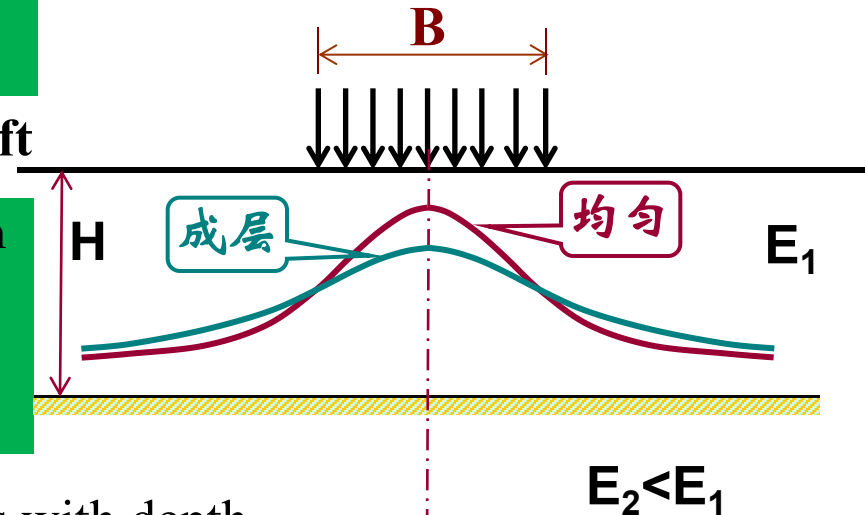
(1) The upper is soft and sublayer is stiffness

- ❖ σ_z near the central axis increases than that in homogeneity condition —**stress concentration**;
- ❖ Degree of stress concentration is relate to Stiffness and thickness of soil layer.



(2) The upper is stiffness and sublayer is soft

- ❖ σ_z near the central axis decreases than that in homogeneity condition —**stress dispersion**;
- ❖ Degree of stress concentration is relate to Stiffness and thickness of soil layer.



(3) The deformation modulus of soil increases with depth —**stress concentration**

3.4 Principle of effective stress

- ✱ **Pore water pressure:** in saturated soils, a pressure is transferred by pore water, u is commonly used to show the pressure.
- ✱ The **property of pore water is the same as normal hydrostatic pressure**, the direction is always perpendicular to the surface of action and pore water pressures at any point in each direction are equal.

$$u = \gamma_w h_w$$

Principle of effective stress

✶ **Effective stress:** a stress only transfer by intergranular contact surface. Only the effective stress can make soils compacted and produced strength.

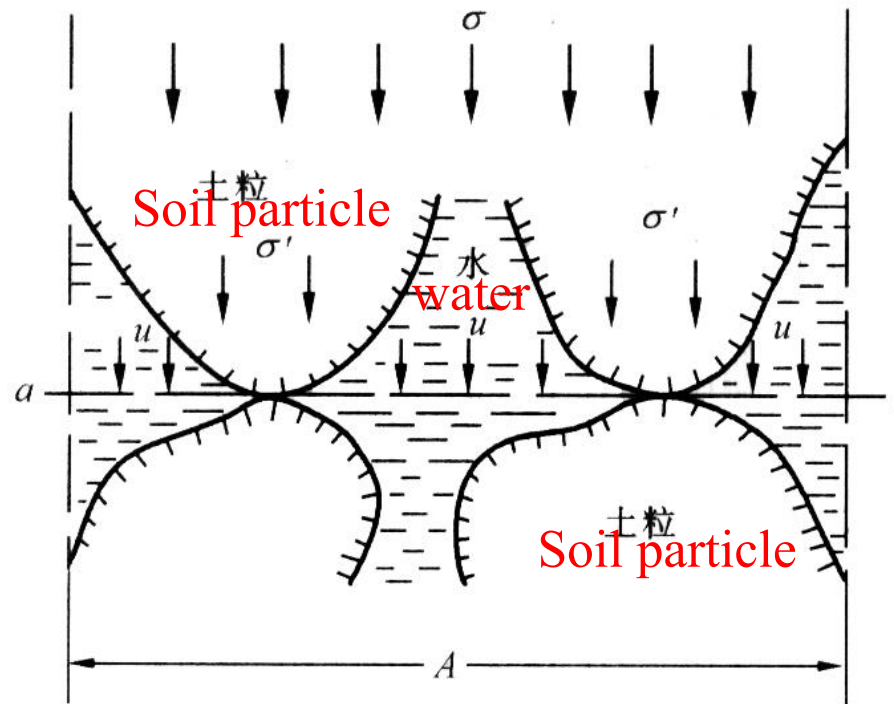
✶ **How to calculate the effective stress?**

✶ It is an average stress that the sum of normal component of all the force on the intergranular contact surface is divided by the total area of the plane.

apostrophe

$$\sigma' = \frac{N_s}{A}$$

Principle of effective stress



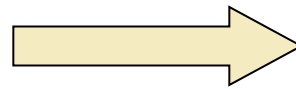
schematic diagram

● Terzaghi's effective pressure principle

✿ For a certain research plane in saturated soil

$$\sigma A = N_s + (A - A_s)u$$

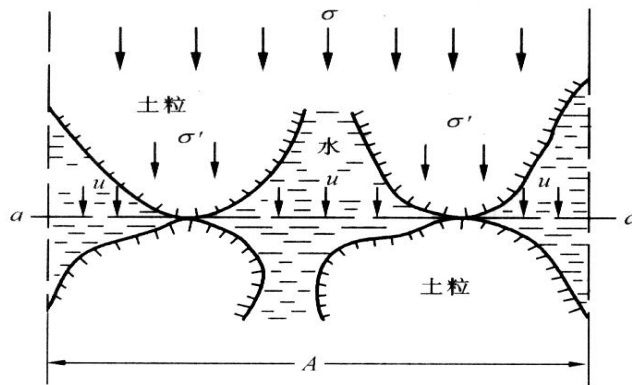
$$\sigma' = \frac{N_s}{A}$$



$$\sigma = \sigma' + (1 - a)u$$



$$\sigma = \sigma' + u$$

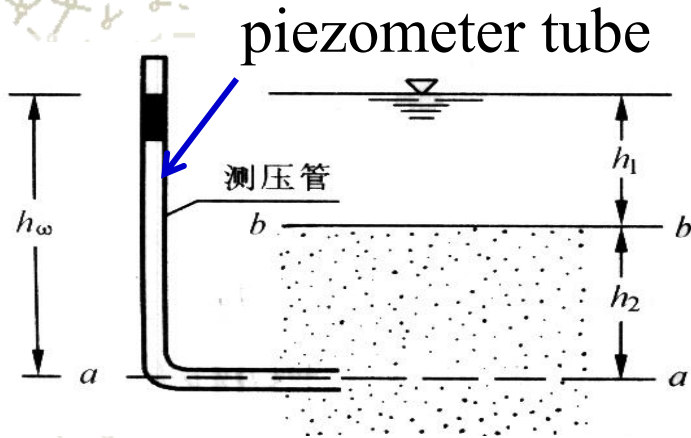


Principle of effective stress

💡 If we keep the total stress at constant, there exists **mutual transformation** between pore water pressure and effective stress. It means that the decrease (**increase**) of pore water pressure is equal to the increase (**decrease**) of effective stress.

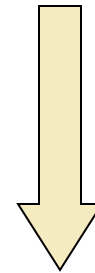
💡 The principle is very important! !

pore water pressure and effective stress under the static water condition



$$\sigma = \gamma_w h_1 + \gamma_{sat} h_2$$

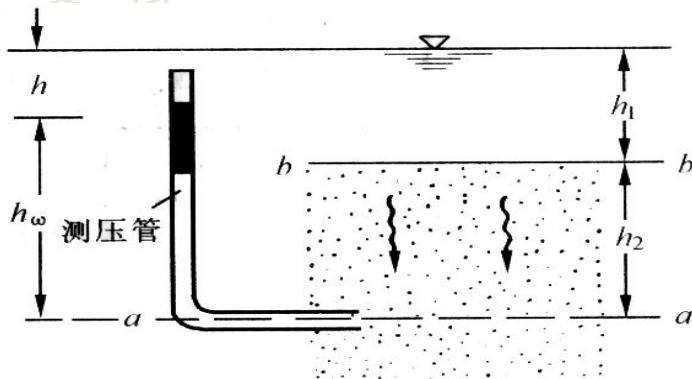
$$u = \gamma_w h_w = \gamma_w (h_1 + h_2)$$



$$\sigma' = \sigma - u = \gamma' h_2$$

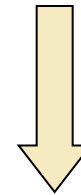
pore water pressure and effective stress under the action of stable seepage

Downward seepage



$$u = \gamma_w h_w = \gamma_w (h_1 + h_2 - h)$$

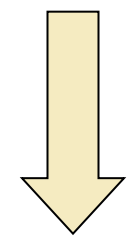
$$\sigma = \gamma_w h_1 + \gamma_{sat} h_2$$



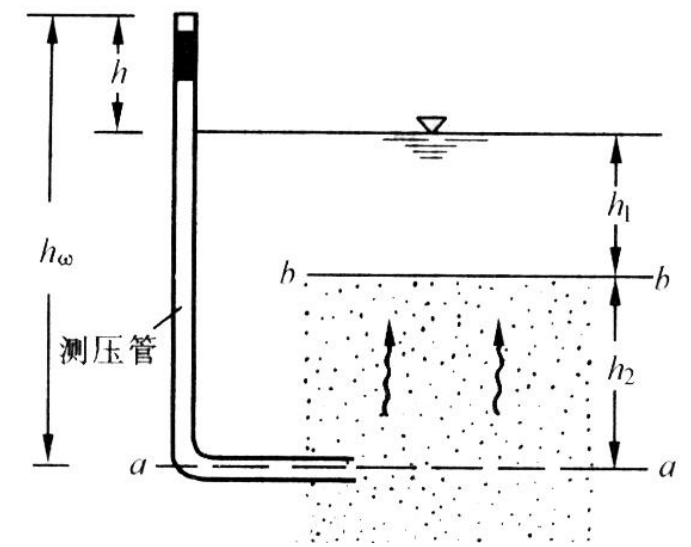
$$\sigma' = \sigma - u = \gamma' h_2 + \gamma_w h$$

$$\sigma = \gamma_w h_1 + \gamma_{\text{sat}} h_2$$

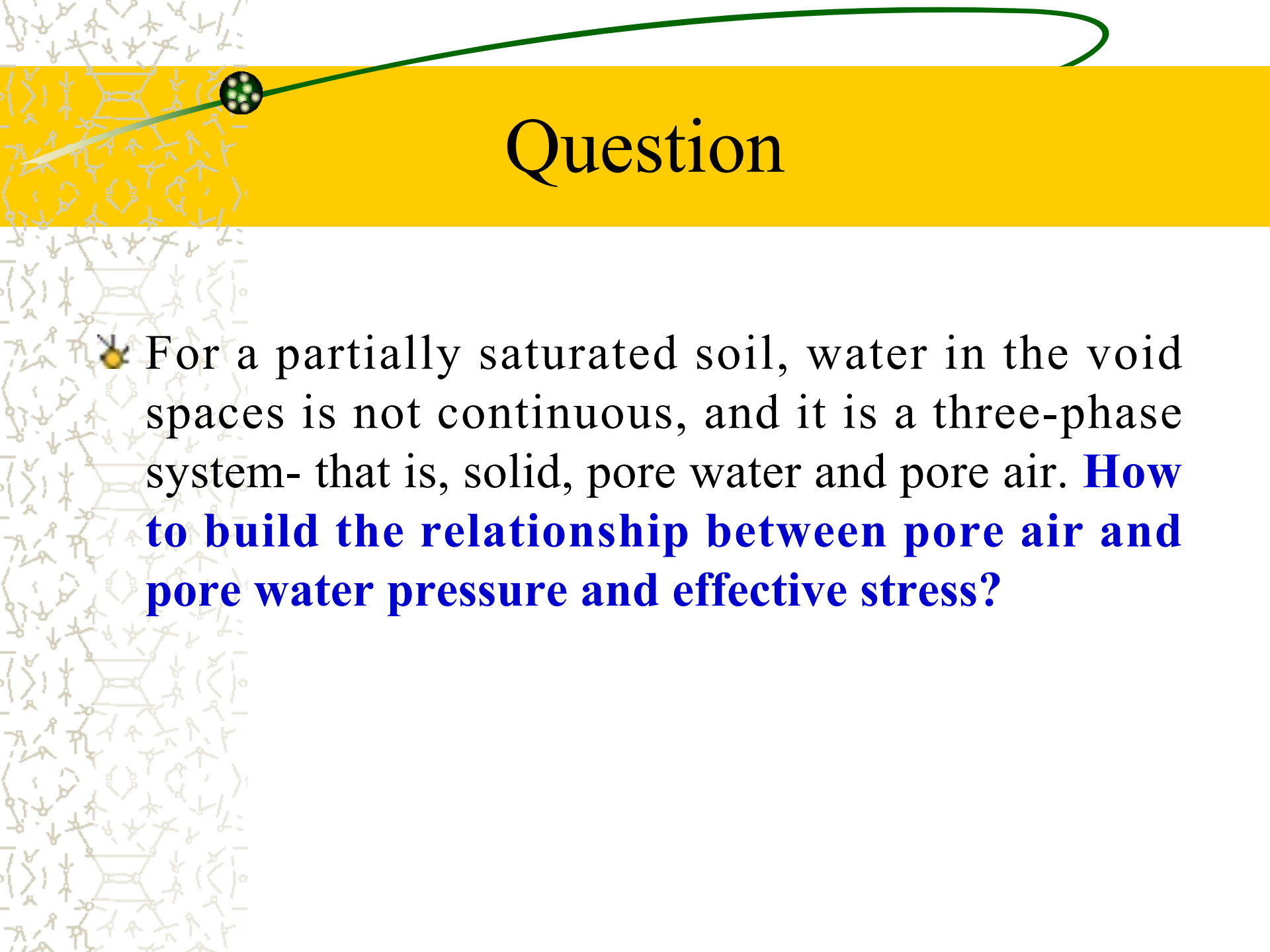
$$u = \gamma_w h_w = \gamma_w (h_1 + h_2 + h)$$



$$\sigma' = \sigma - u = \gamma' h_2 - \gamma_w h$$



$$\gamma' = \gamma_w \frac{h}{h_2} = \gamma_w i = j$$



Question

✱ For a partially saturated soil, water in the void spaces is not continuous, and it is a three-phase system- that is, solid, pore water and pore air. **How to build the relationship between pore air and pore water pressure and effective stress?**

Homework

✱ The plan of a uniformly loaded rectangular area is shown in figure 1. Determine the **vertical stress increases** below point **A** at a depth $z=2$ m.

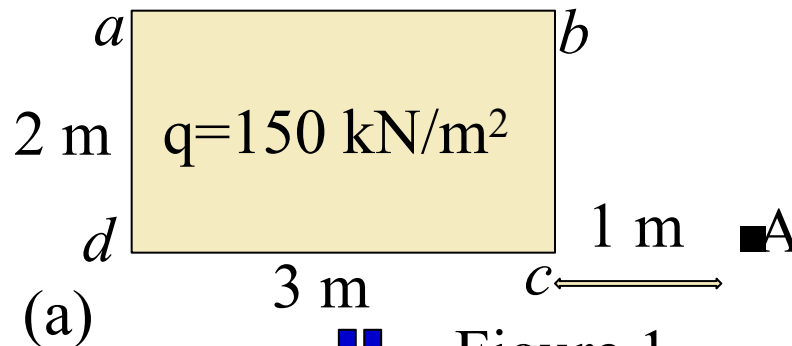


Figure 1

reminder

